Fixyachips Academy | Technical Capabilities Brief

Platform Architecture: 100% Serverless Cloud Infrastructure (AWS)

Market Reach: Global (190+ Countries)

Business Model: High-Margin SaaS (Software as a Service)

Current Production Deployment: Infrastructure Engine v4.2.7

I. Automated Content Infrastructure (The "Media Factory")

Our proprietary media ingestion pipeline explicitly matches our live AWS CloudFormation footprint, eliminating traditional server overhead by deploying a 100% idle-free, event-driven network.

- **Global Video On Demand (VOD) Engine (Phase 1 Live Stack):** Core video assets are housed within encrypted Amazon S3 object storage configured with Intelligent-Tiering logic. Video delivery is accelerated via an global Amazon CloudFront distribution mapping across 600+ edge security points.
- **Neural Translation and Localization Engine:** Automated execution loops leverage neural models to instantly translate structural transcripts, localizing instructional assets into **8 primary languages** natively at the edge network to minimize regional network latency.
- **Generative Knowledge Orchestration (Phase 2 Roadmap Component):** System architecture is engineered to scale seamlessly into **Amazon Bedrock**. In Phase 2, Bedrock foundation model orchestration will natively ingest transcription payloads to automatically recompile field-recorded scripts into production-ready, structured 12-page PDF trade training manual outputs on-demand.

II. Proprietary "Heartbeat" Monitoring & State Management

To mitigate systemic downstream merchant liabilities, eliminate friendly fraud chargebacks, and maintain ironclad audit trails for banking settlement gateways, the system runs an active structural security pulse.

- **State Tracking Loops:** Every student consumption segment generates granular state data points transmitted dynamically across encrypted HTTPS channels. These records are indexed directly to high-throughput, auto-scaling Amazon DynamoDB tables.
- **Defensible Compliance Logging:** Automated validation logic monitors completion rates. Course completion certification maps directly against verifiable, itemized transaction and playback metrics, presenting a fully auditable compliance profile for downstream payment processors.

III. Enterprise-Grade Security, Perimeter Protection & Identity

- **Edge Perimeter Shielding (AWS WAFv2):** Core API and content endpoints are wrapped in an multi-layered Web Access Control List (WebACL) enforcing automated rate limits per source IP

address alongside managed AWS rule groups blocking known malicious injection vectors, bad inputs, and bot patterns.

- **Clean Route Optimization (CloudFront Functions):** Native JavaScript edge routing functions execute runtime uniform resource inspections, ensuring clean URL rewrites directly inside the CloudFront distribution layer without spinning up container dependencies.
- **Identity Provisioning & Data Separation:** Student profiles, authorization profiles, and operational parameters are entirely isolated from external vectors. Internal data movement uses locked IAM roles explicitly requiring end-to-end TLS encryption (`SecureTransport: true`), keeping the platform natively aligned with strict corporate PCI data control layers.

IV. Infinite Financial Scalability

- **Zero-Idle Cost Efficiency:** Infrastructure overhead maintains a direct linear relationship with global usage volume. Total structural baseline operation remains effectively **$0.00** during idle traffic periods, maximizing operating profit margins from day one.